

Sreevaatsav Bavana

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Portfolio website

Personal Profile

Gold medalist in B.Tech Artificial Intelligence at Mahindra University, currently working as Senior Data Scientist at Penguin AI. Strong expertise in Multimodal AI, LLM systems, and Machine Learning, with research interests in learning robotic manipulation policies from multimodal perception, sim-to-real transfer, and grounded, interpretable vision-language models. Passionate about building AI systems that bridge perception, language, and real-world action.

Education

Mahindra University, Hyderabad, India

Aug 2020 - Aug 2024

Bachelor of Technology in Artificial Intelligence

- **Gold Medalist** of the 2020–2024 B.Tech AI batch
- **Cumulative GPA** : 8.76
- **Relevant Courses**: Data Structures and Algorithms, Linear Algebra, Probability and Statistics, Optimization Techniques in AI, DBMS, Machine and Deep Learning, Natural Language Processing, Reinforcement learning

Courses & Certifications

Modern Robotics, Course 1: Foundations of Robot Motion

Apr 2026

Coursera — Northwestern University · Grade: 98.46%

Modern Robotics, Course 2: Robot Kinematics

In Progress

Coursera — Northwestern University

Work Experience

Senior Data Scientist

May 2026 - Present

Penguin AI, Hyderabad, India

- Building end-to-end LLM-based agent pipelines for healthcare workflows, incorporating live production user feedback for self-improving, data-driven agent development.
- Leading AI governance efforts — compliance, bias mitigation, and safety alignment across deployed systems.
- Designing ML system architecture for cost optimization and scalability of agent infrastructure in production.

Data Scientist

Jun 2025 - May 2026

Penguin AI, Hyderabad, India

- Built production-grade LLM workflow systems to automate prior authorization, reducing medical nurse review time by approximately 30-40% with around 80-85% accuracy; currently improving the system with a HITL workflow.
- Working on applied research methodologies to enhance automated medical coding.
- Developed an ML system for appeal denial prediction directly from customer data sources, with a corresponding ETL pipeline.

Associate Data Scientist

Jun 2024 - Jun 2025

Awone AI, Hyderabad, India

- Developed an end-to-end interruptible voice chat application with open-source components with MCP client-server for a user database.
- Improved the inference speed of production LLMs by 15% by quantization and developing speculative decoding n-gram head.
- Built and deployed a BERT-based NER model for healthcare PII redaction using a custom loss function.

Data Science Intern

Jan 2024 - Jun 2024

Awone AI, Hyderabad, India

- Implemented Llama-2 and Mistral from scratch to study their architectures and attention mechanisms, and applied quantization and other optimization techniques to reduce inference latency.
- Finetuned task-specific SLMs for medical-insurance related tasks.

Research intern

Jul 2023 - Sept 2023

Mahindra University, Hyderabad, India

- Surveyed video summarization approaches, challenges, and future directions with a four-person team, laying groundwork for follow-on modeling work.

Publications

WORKSHOP & CONFERENCE PAPERS

GIV-CXR: Densely Grounded, Visually Interpretable, Chest X-Ray Question Answering Dataset
Sreevaatsav Bavana, Adit Rushil Potta, Sai Amrit Patnaik, Nidhi Goyal
MedReasoner Workshop at CVPR, 2026

From Layoff to Reemployment: Analyzing Social Media Narratives
Yayati Gupta, Sreevaatsav Bavana, Sudhira Chegu, Sujith Sai Kalakonda, Yash Mathur, Manmitha Polsani, Sanatan Sukhija
Computational Data and Social Networks (CSoNet 2025), 2025

A Systematic Study on Video Summarization: Approaches, Challenges, and Future Directions
Kajal Kansal, Nikita Kansal, Sreevaatsav Bavana, Bodla Krishna Vamshi, Nidhi Goyal
Proceedings of the 2nd Workshop on User-centric Narrative Summarization of Long Videos, 2023

Projects

Imitation-Learning Policies on a Fairino FR5 Cobot Jun 2026 - Present

- Independent robotics research · Collaborator: Vivek Kanjarla
- Collected ~750 teleoperated demonstrations on a 6-DOF Fairino FR5 across three manipulation tasks, recorded with a wrist-mounted Intel RealSense D405 camera.
 - Built the data pipeline behind them: aligns ~100 Hz robot state logs to 30 Hz camera frames and packages each episode as a LeRobot v3.0 dataset, so every policy trains off one format.
 - Trained ACT, diffusion policy and π_0 / $\pi_{0.5}$ on this data and ran them closed-loop on the arm at 30 Hz, fine-tuning the VLAs with LoRA and NF4/int8 quantization on rented GPUs.
 - π_0 reached 80% on sanding and 65% on colour sorting, each over 20 autonomous trials (320 and 280 demonstrations). $\pi_{0.5}$ is training now for pick-to-light: retrieve an object from a named rack slot and place it.
 - Reimplemented ACT from the paper — CVAE plus transformer, built up from autoencoder and VAE components.

AnimalCLEF 2026: Wildlife Re-identification — 9th Place Apr 2026 - May 2026

- AnimalCLEF 2026 Challenge
- Performed unsupervised clustering of ~2,400 unlabeled images across 4 species (lynx, salamanders, sea turtles, Texas horned lizards).
 - Fused global embeddings (MiewID v3 + MegaDescriptor) with local feature matching (KAZE, ALIKED+LightGlue).
 - Fine-tuned MegaDescriptor with ArcFace loss for a +6.4% gain; used SAM2 to mask background keypoints before matching.
 - Ran 30+ documented experiments; secured 9th place on the leaderboard.

GIV-CXR: Densely Grounded, Visually Interpretable, Chest X-Ray Question Answering Dataset Aug 2024 - Jan 2025

- Supervisors: Dr. Nidhi Goyal, Assistant Professor, Mahindra University
- Curated a densely grounded chest X-ray VQA benchmark with 355,293 question-answer pairs linked to bounding-box annotations across 36 anatomical regions, enabling systematic spatial-reasoning evaluation.
 - Built a hallucination-filtering pipeline combining automated detection with radiologist validation, achieving an 82.4% acceptance rate (18.2% hallucination rejection).
 - Developed grounded VLM baselines; best model achieved 68.1 mIoU for spatial grounding (nearly 2× GPT-4o-mini’s 39.0) while leading on G-Eval answer quality (3.98).

Enhancing video summarization by enhancing text-to-image module Aug 2023 - Dec 2023

- Supervisors: Dr. Nidhi Goyal, Assistant Professor, Mahindra University
- Improved text-to-image projection by augmenting it with a learnable projection layer.
 - Finetuned the selected baseline’s projection module using the QVHighlights dataset.

Achievements

- 9th place** in AnimalCLEF 2026 Wildlife Re-identification Challenge 2026
- 4 years consecutive merit scholarship (2020-2024)** for securing a rank in the top 5% of the branch 2024
- 3rd position** in ICETCI Hackathon (2023), Mahindra University 2023
- Top 150 teams (Rank 111)** in Amazon ML Challenge 2023, a total of 4000+ teams participated 2023

Skills

Languages & Libraries	Python, SQL, Pandas, NumPy, Scikit-learn, HuggingFace, PyTorch, OpenCV, C++
AI Engineering & Infrastructure	LangChain, LangGraph, vLLM, MLFlow, Ray
Cloud & Deployment	AWS, Docker, Async programming, Linux, GitHub Actions, MongoDB, PostgreSQL, Kubernetes
Robot Learning	Imitation learning, ACT, Diffusion Policy, π_0 / $\pi_{0.5}$ VLAs, LoRA & quantized fine-tuning
Robotics & Simulation	ROS 2, LeRobot, Fairino SDK, SO-101, RealSense, MuJoCo, CoppeliaSim

Extra-Curricular

Power-lifting Volunteer for AERO

Coordinated events at Aero 2023 Sports Fest

College Gym committee member (MU)

Oversaw equipment and mentored new students

Yoga club member and volunteer (MU)

Volunteered for the University's celebration of International Yoga Day.